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Vehicle having drone docking station

Abstract

A vehicle that includes a docking station positioned within the interior of the vehicle, and a drone that is configured to mate with the docking station. The vehicle includes an electronic control unit (ECU) that is located within the interior of the vehicle and is configured to communicate instructions to a docking station controller to instruct the drone to disengage from the docking station and perform flight operations. Additionally, the vehicle ECU is configured to communicate instructions to the docking station controller to instruct the drone to cease flight operations and return to and mate with the docking station.

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Background/Summary

FIELD

(1) The present disclosure relates to a vehicle having a drone docking station.

BACKGROUND

(2) This section provides background information related to the present disclosure which is not necessarily prior art.

(3) Drones are used in a wide range of applications such as aerial photography, surveillance, delivery, and inspection. Drones can be used in a variety of environments, including urban, rural, and remote areas. Drones have limited battery life, however, and the range of the drone is limited by this battery life. Moreover, drones may need a flat, safe takeoff and landing area, which may not be readily available in certain areas.

SUMMARY

(4) This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

(5) According to a first aspect of the present disclosure, there is provided a vehicle that includes a

vehicle body defining an interior. A vehicle electronic control unit (ECU) is located within the interior of the vehicle. A docking station is positioned within the interior of the vehicle and includes a docking station controller configured to communicate with the vehicle ECU. A drone is configured to mate with the docking station that is configured to fly. The drone includes a drone controller configured to communicate with the docking station. The vehicle ECU is configured to communicate instructions to the docking station controller to instruct the drone to disengage from the docking station and perform flight operations. The vehicle ECU is further configured to communicate instructions to the docking station controller to instruct the drone to cease flight operations and return to and mate with the docking station.

(6) According to the first aspect, the drone includes a battery, and the docking station is configured to electrically charge the battery when the drone is mated with the docking station.

(7) According to the first aspect, the vehicle comprises a plurality of sensors that communicate various information to the vehicle ECU, and the vehicle ECU is configured to communicate the information to the docking station controller.

(8) According to the first aspect, the docking station controller is configured to determine whether to permit the drone to disengage from the docking station based on the information provided by the sensors and communicated from the vehicle ECU, and determine whether to instruct the drone to cease flight operations and return to and mate with the docking station based on the information provided by the sensors and communicated by the vehicle ECU.

(9) According to the first aspect, the vehicle ECU is configured to communicate with a network and receive weather information from the network, and communicate the weather information to the docking station controller.

(10) According to the first aspect, the docking station controller is configured to determine whether to permit the drone to disengage from the docking station based on the weather information and communicated from the vehicle ECU, and determine whether to instruct the drone to cease flight operations and return to and mate with the docking station based on the weather information and communicated by the vehicle ECU.

(11) According to the first aspect, the drone controller is configured to communicate with a remote operating device.

(12) According to the first aspect, the vehicle ECU is configured to communicate with the remote operating device.

(13) According to the first aspect, the vehicle includes a graphic user interface (GUI) that communicates with the vehicle ECU, the GUI being configured to be manipulated by an occupant of the vehicle to instruct the vehicle ECU to communicate with the docking station controller to instruct the drone to perform the flight operations or cease the flight operations.

(14) According to the first aspect, the interior includes a cargo bin having a movable lid to provide and prevent access to the cargo bin, the docking station being located in the cargo bin.

(15) According to the first aspect, the vehicle body includes an opening that is configured to permit the drone to enter and exit the vehicle.

(16) According to the first aspect, the opening is a sun roof or a moon roof.

(17) According to the first aspect, the vehicle includes a removable roof, and when the roof is removed from the vehicle the drone is permitted to exit and enter the vehicle.

(18) According to a second aspect of the present disclosure, there is provided a method for operating a drone using the vehicle described above. The method may include receiving, by the vehicle ECU, a request to initiate the flight operations from either a graphic user interface (GUI) of the vehicle or a remote operating device. Based on the request received by the vehicle ECU to initiate the flight operations, the ECU is configured to communicate with vehicle sensors and/or a network in communication with the ECU to determine whether weather conditions are suitable to conduct the flight operations, and based on signals generated by the vehicle sensors that are indicative of the weather conditions and/or information received from the network that are

indicative of the weather conditions, the vehicle ECU is configured to communicate instructions to the docking station controller either to permit the drone to conduct the flight operations or prevent the drone from conducting the flight operations.

(19) According to the second aspect, after the vehicle ECU receives the request to initiate the flight operations, the method further comprises determining whether the vehicle is in motion and, based on a determination whether the vehicle is in motion, the vehicle ECU is configured to communicate instructions to the docking station controller either to permit the drone to conduct the flight operations or prevent the drone from conducting the flight operations.

(20) Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS

(1) The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations, and are not intended to limit the scope of the present disclosure.

(2) FIG. 1 is an overhead view of a vehicle according to a principle of the present disclosure;

(3) FIG. 2 is an overhead view of an interior of the vehicle illustrated in FIG. 1;

(4) FIG. 3 is a perspective view of an example drone docking station having a drone mounted thereto, located within the interior of the vehicle illustrated in FIG. 2;

(5) FIG. 4 is a schematic diagram depicting a vehicle system including a drone docking system according to a principle of the present disclosure;

(6) FIG. 5 is a flowchart depicting an example method for controlling a drone using the drone docking system of the present disclosure;

(7) FIG. 6 is a flowchart depicting another example method for controlling a drone using the drone docking system of the present disclosure; and

(8) FIG. 7 is a flowchart depicting another example method for controlling a drone using the drone docking system of the present disclosure.

(9) Corresponding reference numerals indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

(10) Example embodiments are provided so that this disclosure will be thorough, and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.

(11) FIG. 1 illustrates an example vehicle **100** that may include a drone docking station **304** (FIG. 3) according to the present disclosure. Vehicle **100** has a body **102** defining a roof **104**. In the illustrated embodiment, roof **104** may define an opening **106**. Opening **106** may include, for example, a glass panel **108** defining a sun roof, a moon roof, or a panoramic roof that permits light to enter the vehicle. Panel **108** does not necessarily need to be formed of glass or a transparent material. In contrast, panel **108** may be an opaque removable roof panel. In either case, it should be understood that panel **108** is movable to provide access to an interior **200** (FIG. 2) of vehicle **100**.

(12) FIG. 2 illustrates an example interior **200** of vehicle **100**. In the illustrated example, interior **200** includes a front row of seats **202**, a second row of seats **204**, and a third row of seats **206**. It

should be understood, however, that second and third rows of seats **204** and **206** may be omitted such that vehicle includes a rear cargo storage area in place thereof. Interior **200** may also include a graphic user interface (GUI) **208** on a dashboard **210** of vehicle **100** that is in communication with an electronic control unit **402** (FIG. 4) of vehicle **100**. Interior **200** may include a cargo bin **211** located proximate GUI **208** between the seats **212** of first row **202**.

(13) As best shown in FIG. 3, cargo bin **211** may have a movable lid **306** that may be opened to expose and provide access to a storage area **308** of cargo bin **202**, and closed to enclose and prevent access to storage area **308**, as is known in the art. In accordance with the present disclosure, storage area **308** may include a docking station **304** configured to mate with a drone **302**. As will be described in more detail later, lid **306** may be opened to permit drone **302** to detach from docking station **304** and exit the storage area **308** while being controlled by an operator. Alternatively, the operator may simply open lid **306**, manually detach drone **302** from docking station **304**, manually remove drone **302** from vehicle **100**, and then operate drone **302**. If operator elects to control drone **302** remotely to exit storage area **308** and vehicle **100**, the operator may first open or remove panel **108** located on roof **104** of vehicle **100** to permit drone **302** to exit vehicle **100** via flight.

(14) Now referring to FIG. 4, a schematic representation of a vehicle system **400** for controlling drone **302** that includes drone docking station **304** is illustrated. In addition to docking station **304**, system **400** may include ECU **402**, GUI **208**, a communication network **404**, various vehicle sensors **406**, and an optional remote operating device **408**. Example remote operating devices **408** include remote controllers, mobile phones, portable or desktop computers, electronic tablets, and the like.

(15) ECU **402** includes a communication module **410** for communicating with various modules located throughout vehicle **100** that control various features of the vehicle such as, for example, climate control modules (not shown), vehicle exterior and interior lighting modules (not shown), engine control modules (not shown), window control modules (not shown), and the like. Communication module **410** also enables ECU **402** to communicate with GUI **208**, docking station **304**, and network **404**. ECU **402** also includes a memory **411** such as a RAM or ROM, and a controller **412** that communicates with communication module **410**, wherein controller **412** is configured to communicate various instructions to GUI **208** and docking station **304** (as well as other vehicle modules like those described above) via communication module **410**.

(16) ECU **402** is also in communication with vehicle sensors **406**. Example vehicle sensors **406** that may communicate with ECU **402** include interior and exterior temperature sensors (not shown), rain sensors (not shown), exhaust gas sensors (not shown), engine load sensors (not shown), oxygen sensors (not shown), fuel level sensors (not shown), and other vehicle sensors known to one skilled in the art. Specific sensors **406** that may communicate with ECU **402** to assist with operating drone **302** include rain sensors that are configured to transmit signals indicative of whether vehicle **100** is being exposed to rain, velocity sensors that are configured to transmit signals indicative of a velocity of the vehicle **100**, wiper motor sensors that are configured to transmit signals indicative of the vehicle windshield wiper blades (not shown) being operated, and accelerometer or gyro-sensors that are configured to transmit signals indicative of an acceleration of the vehicle **100** or whether the vehicle **100** is located on a slope (i.e., not located on flat ground). These sensors **406** are important because these sensors **406** can generate signals that assist in determining whether to deploy or recall drone **302**, or assist in operating drone **302** when it enters or exits vehicle **100**.

(17) For example, rain sensors can communicate signals indicative of whether it is raining. Operation of drone **302** may not be desirable in adverse weather conditions. Thus, based on a signal indicative of whether it is raining transmitted by the sensor **406**, ECU **402** may be configured to send an instruction to docking station **304** that either prevents drone **302** from being deployed in the adverse weather conditions, or send an instruction to docking station **304** to recall a deployed drone **302** back to the docking station **304** so as to avoid drone **302** from being operated and/or

damaged by the adverse weather conditions. Windshield wiper motor sensors **406** may be used in the same manner. These operations will be described in more detail later.

(18) It also may not be desirable to deploy drone **302** from vehicle **100**, or recall drone **302** back to vehicle **100**, when vehicle **100** is in motion. Accordingly, signals indicative of the velocity of vehicle **100** or an acceleration of vehicle **100** that are transmitted to ECU **402** can be used to assist in operation of drone **302**. For example, if ECU **402** receives a signal indicative of the velocity or acceleration of the vehicle **100**, ECU **402** may communicate an instruction to docking station **304** that prevents drone **302** from being deployed from docking station **304** while the vehicle **100** is in motion. Conversely, if drone **302** is already deployed outside of vehicle **100** and the vehicle **100** is in motion, ECU **402** can send an instruction to docking station **304**, which is subsequently communicated to drone **302**, that prevents drone **302** from returning to docking station **304**. In this regard, it is much more difficult to return drone **302** to the docking station **304** when the vehicle **100** is in motion and such an instruction can prevent drone **302** from attempting to dock with docking station **304** while the vehicle **100** is in motion to prevent unnecessary damage to either vehicle **100** or drone **302** in such an instance.

(19) Docking station **304** includes a docking station communication module **414** that communicates with ECU **402** and network **404**, a charging module **416** that is configured to charge a battery **418** of drone **302** when drone **302** is mated with docking station **304**, a docking station controller **420** that is configured to send instructions to drone **302** (via network **400** when drone **302** is deployed, or directly to drone **302** via communication module **414** when drone **302** is mated with docking station **304**) and to charging module **416**, and a GPS module **600**. Docking station controller **420** may send instructions to charging module **416** to begin charging drone **302** when drone **302** is mated with docking station **304**, or send instructions to charging module **416** to cease charging drone **302** upon receive of a signal indicative that battery **418** of drone **302** has fully charged.

(20) GUI **208** communicates with ECU **402** and may be a touch-screen operated GUI **208**. As is known in the art, GUI **208** can be used to operate various systems of vehicle **100**. For example, an occupant of vehicle **100** can operate a climate control system or a radio of the vehicle **100** using GUI **208** by touching various icons **500** on a screen **510** of the GUI **208**. Drone **302** may be operated from GUI **208** in a similar manner. For example, if an occupant of vehicle **100** elects to deploy drone **302** from the docking station **304**, GUI **208** may include various icons **500** that may be touched to operate drone **302** as best shown in FIG. **6**. For example, a “deploy drone” icon **500** may be located on GUI **208** and upon the icon **500** being touched by the occupant, a signal may be sent from GUI **208** to ECU **402**, which then sends a signal to docking station **304** to deploy the drone **302**.

(21) GUI **208** may also be configured to display information relative to drone **302** such as, for example and without limitation, whether drone **302** is connected to docking station **304** or whether battery **418** of drone **302** is charged (e.g., by displaying a percentage of charge remaining in battery **418**). If drone **302** is in use outside of vehicle **100**, GUI **208** may also be used to control movement of drone **302** (e.g., an elevation of flight, velocity of drone **302**, and the like). Any instruction input into GUI **208** is communicated to ECU **402**, which is then communicated to docking station **304**, which is then communicated to drone **302** via network **404**. An example type of GUI **208** is a Uconnect® system.

(22) Network **404** may be a wireless communication network operating in accordance with one or more standards including, but not limited to, Wi-Fi, 3GPP (3rd Generation Partnership Project), 4GPP (4th Generation Partnership Project), 5GPP (5th Generation Partnership Project), LTE (long term evolution), LTE Advanced, RFID, IEEE 802.11, Bluetooth, AMPS (advanced mobile phone services), digital AMPS, GSM (global system for mobile communications), CDMA (code division multiple access), LMDS (local multi-point distribution systems), MMDS (multi-channel-multi-point distribution systems), V2V (vehicle-to-vehicle), V2X (vehicle-to-anything), and/or variations

thereof. In the illustrated embodiment, network **404** may independently communicate with drone **302**, docking station **304**, ECU **402**, and remote operating device **408**. Network **404** may also serve as a wireless conduit of communication between these devices.

(23) Drone **302** includes a drone communication module **422**, a drone controller **424**, battery **418**, various sensors **426**, a memory **431** such as a RAM or ROM, and a GPS module **700**. Optional components include a camera **428**, a light-emitting device **430** that can be operated in a manner similar to a vehicle's headlights (e.g., to illuminate the ground beneath drone **302** during flight operations), and a depth device **432** configured to determine a depth of a puddle or other body of water. For example, depth device **432** can emit a laser, sonar, or some other type of energy that can assist in determining a depth).

(24) Drone communication module **422** is configured to receive and transmit information from network **400** to other elements of the system such as, for example, docking station **304** and remote operating device **408**. If drone **302** is mated with docking station **304**, drone communication module **422** may receive and transmit information directly to docking station control module **414**.

(25) Drone sensors **426** may be similar to vehicle sensors **406**. Example drone sensors **426** include an altimeter that generates a signal indicative of an altitude of the drone **302**, a GPS sensor that generates a signal indicative of a location of the drone **302** during flight operations, a velocity sensor that generates a signal indicative of a velocity of the drone **302** during flight operations, a rain sensor that generates a signal indicative of whether it is raining, a battery sensor that generates a signal indicative of a percentage of battery power remaining in battery **418**, accelerometers that generate signals indicative of an acceleration of drone **302**, and gyro-sensors that generate signals indicative of an orientation of drone **302**. These signals may be transmitted to drone controller **424**, which are then sent to communication module **422** to be transmitted to other elements of the system **400** via network **404**.

(26) If drone **302** includes a camera **428**, images or video footage captured by camera **428** may be transmitted via communication module **422** to vehicle **100** or remote operating device **408**. For example, any images or footage captured by camera **428** may be communicated to vehicle **100** (i.e., to ECU **402** via network **400**, or to docking station **304** via network **400** and then to ECU **402** from the docking station **304**) to be displayed by GUI **208** on screen **510**. Alternatively, if remote operating device **408** includes a display **510**, the images or footage captured by camera **428** transmitted to remote operating device **408** via network **404** may be displayed by remote operating device **408**. In another alternative, the images and/or video obtained by camera **428** may be stored in memory **431** to be downloaded and viewed later.

(27) Remote operating device **408** may include a remote operating device communication module **434**, a GPS module **435**, and a remote operating device controller **436**. Remote operating device communication module **434** permits remote operating device **408** to communicate with network **404** and other elements of system **400** such as drone **302**, docking station **304**, and ECU **402**. Remote operating device controller **436** may include a drone operating system software stored therein for operating drone **302**. Based on inputs received from an operator, remote operating device controller **436** may then communicate instructions to any of drone **302**, docking station **304**, and ECU **402** via remote operating device communication module **434**.

(28) For example, if someone using remote operating device **408** desires to deploy drone **302** from docking station **304**, the operator may use remote operating device **408** to send an instruction to docking station **304** via network **404**, which then communicates with drone **302** to initiate deployment of drone **302** from docking station **304** (in such a case, it is assumed that panel **108** (FIG. **1**) is open or removed).

(29) In another embodiment, the instructions sent from remote operating device **408** to docking station **304** via network **404** may be further communicated to ECU **402**. Upon receipt of such an instruction by ECU **402**, ECU **402** may communicate with a vehicle sensor **406** such as, for example, a sensor **406** that generates a signal indicative of whether panel **108** is open or removed)

to determine whether it is safe for drone **302** to exit the vehicle **100** through the opening **106** formed in roof **104** of vehicle **100**. After ECU **402** receives a signal indicative of panel **108** being open or removed so as to not obstruct opening **106**, ECU **402** may then communicate a signal back to docking station **304** that permits drone **302** to be deployed an exit vehicle **100** through opening **106**.

(30) In other embodiments, ECU **402** may also communicate with other sensors **406** or network **404** to determine whether weather conditions are such that drone **302** may be deployed. For example, if sensor **406** is a rain sensor that communicates a signal indicative of the vehicle **100** being exposed to rain to ECU **402**, ECU **402** may transmit an instruction to docking station **304** that prevents drone **302** from being deployed. Alternatively, ECU **402** may communicate with network **404** to retrieve current weather conditions. Based on the information regarding the current weather conditions received from network **404**, ECU **402** is configured to send an instruction to docking station **304** that either permits or prevents drone **302** from deploying from docking station **304**.

(31) Drone **302** may be any type of drone known to one skilled in the art, provided that drone **302** is sized to fit within vehicle **100**. Examples of types of drones include, but are not limited to, a helicopter drone, a fixed-wing drone, a multi-rotary-wing drone, a blimp drone, a flexible-wing drone, and the like. Drone **302** may be capable of both autonomous navigation and operating via remote control.

(32) Now referring to FIGS. **5** and **6**, a method of operating drone **302** will be described. As noted above, drone **302** may be operated using GUI **208** or remote operating device **408**. Description of using GUI **208** to operate drone **302** will first be described. Firstly, if docking station **304** and drone **302** are located within cargo bin **202**, lid **306** must be opened (step **1000**) by an occupant of vehicle **100** (this step may be omitted if, for example, docking station **304** is located in a rear storage area of vehicle **100** (i.e., if vehicle **100** does not include second or third rows **204**, **206** of seats and instead includes a rear storage area)).

(33) After opening lid **305**, an occupant may then use GUI **208** to initiate flight operations of drone **302** (step **1010**). At this time, an occupant of vehicle **100** may input information into GUI **208** such as, for example, a destination for drone **302** to travel, a desired flight path, a desired altitude for the flight, a time period of the flight, and the like by interacting with various icons **500** (FIG. **6**) on a screen **510** of the GUI. At this time, GUI **208** communicates a signal to ECU **402** that drone **302** is to be deployed (step **1020**) and that includes the desired information input to the GUI **208** by the occupant.

(34) Upon receipt of this signal and information, ECU **402** communicates with various sensors **406** and/or network **404** to determine whether conditions are acceptable to initiate deployment of drone **302** (step **1030**). These conditions include obtaining information relative to current weather conditions from network **404**, or obtaining signals from sensors **406** that are indicative of the weather that vehicle **100** is currently being exposed (i.e., a signal indicative of rain from rain sensors or a signal indicative of the vehicle wiper blades being operated, for example). Other conditions include whether the vehicle **100** is in motion, whether the vehicle **100** is at an orientation (i.e., on a slope) that would make it difficult to deploy drone **302** from vehicle **100**, and whether panel **108** is open or removed to permit drone **302** to exit the vehicle **100** during flight. If these conditions are satisfactory, ECU **402** may transmit an instruction (step **1040**) to docking station **304** to permit drone **302** to begin flight operations, detach from docking station **304**, and exit vehicle **100** through opening **106** provided in roof **104** of vehicle **100**.

(35) Upon receipt by docking station **304** of instructions from ECU **402**, docking station **304** communicates instructions to drone **302** to initiate flight and including the desired flight characteristics input into GUI **208** by the occupant (step **1050**). Upon receipt of these instructions, drone **302** may autonomously detach from docking station **304** and exit vehicle **100** through opening **106** (step **1060**). At this time, drone **302** may continue to operate autonomously using the

desired flight characteristics input by the occupant to GUI **208** before returning to docking station **304**.

(36) Alternatively, drone **302** may autonomously detach from docking station **304** and exit vehicle **100** before operation of drone **302** is transferred back to GUI **208** (step **1070**). In such a case, drone **302** may exit vehicle **100** and reach a predetermined altitude before operation of drone **302** is transferred back to GUI **208**. Then, the occupant can use various control icons on GUI **208** to move in a particular direction, move at a desired velocity, reach a desired altitude, use camera **428**, operate light source **430**, operate depth detection device **432**, and the like (step **1080**). At this time, the instructions input into GUI **208** are communicated to ECU **402**, which then communicates the instructions to docking station **304**, which communicates the instructions to drone **302** via network **404**.

(37) If drone **302** is operating autonomously based on instructions input to GUI **208** and the desired flight time has elapsed, drone **302** may then automatically return to vehicle **100**, pass through opening **106**, and mate with docking station **304** (step **1090**). Alternatively, if drone **302** is being manually operated, in step **1090** the occupant can input instructions into GUI **208** that direct drone **302** back to vehicle **100**. The occupant can operate drone **302** to reenter vehicle **100** and mate with docking station **304**, or the occupant can operate drone **302** to return to a location exterior to vehicle **100** before manually retrieving drone **302** and manually mating drone **302** with docking station **304**. It should be understood that if vehicle **100** has moved during flight operations, a location of vehicle **100** can be transmitted to drone **302** via network **404**. In this regard, both drone **302** and docking station **304** may include a GPS module **600**, **700** that may coordinate with each other.

(38) It should be understood that during flight operations, ECU **402** may continually communicate with network **404** and/or sensors **406** to continually make a determination whether conditions are satisfactory for continuing flight operations (e.g., continually checking weather information). If for whatever reason ECU **402** determines that flight operations should cease, ECU **402** may send an instruction to docking station **304** to have drone **302** return to vehicle **100**, which is then communicated to drone **302** using network **404**. Moreover, it should be understood that information obtained from drone sensors **426** can be continuously transmitted from drone **302** to docking station **304** and ECU **402** via network **404**. If drone sensors **426** generate signals indicative of poor operation (e.g., motor issues, poor flight conditions, and the like), ECU **402** can transmit instructions to docking station **304** that are transmitted to drone **302** via network **404** to cease flight operations and return to vehicle **100**.

(39) A similar process can be used if drone **302** is being operated by remote operating device **408**. It should be understood, however, that use of remote operating device **408** enables an operator of the remote operating device **408** to deploy drone **302** from vehicle **100** if the operator is located at a location remote from vehicle **100**. For example, referring to FIG. 7, a control process of operating drone **302** when the operator is located remote from vehicle **100** and plans to control drone **302** using remote operating device **408** is illustrated. In the process of FIG. 7, it should be understood that operator is not located in vehicle **100** when operating drone **302**.

(40) First, in step **2000**, operator may open up a software program stored on remote operating device **408** that is configured to operate drone **302**. After opening the software program, a screen **510** of the remote operating device **408** can display an interface similar to that of GUI **208** (see, e.g., FIG. 6) and, the operator may input various information such as, for example, a destination for drone **302** to travel, a desired flight path, a desired altitude for the flight, a time period of the flight, and the like by interacting with various icons **500** (step **2010**; FIG. 6). At this time, remote operating device **408** communicates a signal to either docking station **304** and/or ECU **402** via network **404** that drone **302** is to be deployed (step **2020**) and that includes the desired information input to the remote operating device **408** by the operator.

(41) Upon receipt of this signal and information, ECU **402** communicates with various sensors **406**

and/or network **404** to determine whether conditions are acceptable to initiate deployment of drone **302** (step **2030**). These conditions include obtaining information relative to current weather conditions from network **404**, or obtaining signals from sensors **406** that are indicative of the weather that vehicle **100** is currently being exposed (i.e., a signal indicative of rain from rain sensors or a signal indicative of the vehicle wiper blades being operated, for example). Other conditions include whether the vehicle **100** is in motion, whether the vehicle **100** is at an orientation (i.e., on a slope) that would make it difficult to deploy drone **302** from vehicle **100**, and whether panel **108** is open or removed to permit drone **302** to exit the vehicle **100** during flight. If these conditions are satisfactory, ECU **402** may transmit an instruction (step **2040**) to docking station **304** to permit drone **302** to begin flight operations, detach from docking station **304**, and exit vehicle **100** through opening **106** provided in roof **104** of vehicle **100**. If panel **108** is obstructing opening **106**, ECU **402** can instruct a motor (not shown) that moves panel **108** to move panel **108** so as to not obstruct opening **106** (i.e., controller **402** can open the sunroof, moonroof, or panoramic roof panel).

(42) Upon receipt by docking station **304** of instructions from ECU **402**, docking station **304** communicates instructions to drone **302** to initiate flight and including the desired flight characteristics input into remote operating device **408** by the operator (step **2050**). Upon receipt of these instructions, drone **302** may autonomously detach from docking station **304** and exit vehicle **100** through opening **106** (step **2060**). At this time, drone **302** may continue to operate autonomously using the desired flight characteristics input by the operator to remote operating device **408** before returning to docking station **304**. Alternatively, drone **302** may autonomously detach from docking station **304** and exit vehicle **100** before operation of drone **302** is transferred back to remote operating device (ROD) **408** (step **2070**). In such a case, drone **302** may exit vehicle **100** and reach a predetermined altitude before operation of drone **302** is transferred back to remote operating device **408**. Then, the occupant can use various control icons on the display of remote operating device **408** to move in a particular direction, move at a desired velocity, reach a desired altitude, use camera **428**, operate light source **430**, operate depth detection device **432**, and the like (step **2080**). At this time, the instructions input into remote operating device **408** are communicated to ECU **402**, which then communicates the instructions to docking station **304**, which communicates the instructions to drone **302** via network **404**.

(43) If drone **302** is operating autonomously based on instructions input to remote operating device **408** and the desired flight time has elapsed, drone **302** may then automatically return to vehicle **100**, pass through opening **106**, and mate with docking station **304** (step **2090**). Alternatively, if drone **302** is being manually operated by remote operating device **408**, in step **2090** the operator can input instructions into remote operating device **408** that direct drone **302** back to vehicle **100**. The operator can operate drone **302** to reenter vehicle **100** and mate with docking station **304**, or the occupant can operate drone **302** to return to the operator's location exterior to vehicle **100** before manually retrieving drone **302** and manually mating drone **302** with docking station **304** when desired. It should be understood that if vehicle **100** has moved during flight operations, a location of vehicle **100** can be transmitted to drone **302** via network **404**. In this regard, both drone **302** and docking station **304** may include a GPS module that may coordinate with each other.

(44) It should be understood that during flight operations, ECU **402** may continually communicate with network **404** and/or sensors **406** to continually make a determination whether conditions are satisfactory for continuing flight operations (e.g., continually checking weather information). If for whatever reason ECU **402** determines that flight operations should cease, ECU **402** may send an instruction to docking station **304** to have drone **302** return to vehicle **100**, which is then communicated to drone **302** using network **404**. Moreover, it should be understood that information obtained from drone sensors **426** can be continuously transmitted from drone **302** to docking station **304** and ECU **402** via network **404**. If drone sensors **426** generate signals indicative of poor operation (e.g., motor issues, poor flight conditions, and the like), ECU **402** can transmit

instructions to docking station **304** that are transmitted to drone **302** via network **404** to cease flight operations and return to vehicle **100**.

(45) The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. A vehicle, comprising: a vehicle body defining an interior that includes a front row of seats including a driver's seat and at least one passenger seat, wherein a cargo bin is located between the driver's seat and the at least one passenger seat that has a movable lid to provide and prevent access to the cargo bin; a vehicle electronic control unit (ECU) located within the interior of the vehicle; a docking station positioned within the cargo bin and including a docking station controller configured to communicate with the vehicle ECU; and a drone configured to mate with the docking station, the drone being configured to fly and including a drone controller configured to communicate with the docking station; wherein the vehicle ECU is configured to communicate instructions to the docking station controller to instruct the drone to disengage from the docking station and perform flight operations, and wherein the vehicle ECU is configured to communicate instructions to the docking station controller to instruct the drone to cease flight operations and return to and mate with the docking station.
2. The vehicle according to claim 1, wherein the drone includes a battery; and the docking station is configured to electrically charge the battery when the drone is mated with the docking station.
3. The vehicle according to claim 1, wherein the vehicle comprises a plurality of sensors that communicate various information, including weather information, to the vehicle ECU, and the vehicle ECU is configured to communicate the information to the docking station controller.
4. The vehicle according to claim 3, wherein the docking station controller is configured to determine whether to permit the drone to disengage from the docking station based on the information provided by the sensors and communicated from the vehicle ECU, and determine whether to instruct the drone to cease flight operations and return to and mate with the docking station based on the information provided by the sensors and communicated by the vehicle ECU.
5. The vehicle according to claim 3, wherein the vehicle ECU is further configured to: determine, based on the weather information received from the plurality of sensors, that it is currently not raining; send, based on the determination that it is not currently raining, instructions to perform flight operations to the drone controller.
6. The vehicle according to claim 3, wherein the vehicle ECU is further configured to: determine, based on the information received from the plurality of sensors, that the vehicle is stationary; and send, based on the determination that the vehicle is stationary, instructions to perform flight operations to the drone controller.
7. The vehicle according to claim 1, wherein the vehicle ECU is configured to communicate with a network and receive weather information from the network, and communicate the weather information to the docking station controller.
8. The vehicle according to claim 7, wherein the docking station controller is configured to determine whether to permit the drone to disengage from the docking station based on the weather information communicated from the vehicle ECU, and determine whether to instruct the drone to cease flight operations and return to and mate with the docking station based on the weather information and communicated by the vehicle ECU.

9. The vehicle according to claim 1, wherein the drone controller is configured to communicate with a remote operating device.
 10. The vehicle according to claim 9, wherein the vehicle ECU is configured to communicate with the remote operating device.
 11. The vehicle according to claim 1, wherein the vehicle includes a graphic user interface (GUI) that communicates with the vehicle ECU, the GUI being configured to be manipulated by an occupant of the vehicle to instruct the vehicle ECU to communicate with the docking station controller to instruct the drone to perform the flight operations or cease the flight operations.
 12. The vehicle according to claim 11, wherein the GUI further includes: a status element configured to display a drone docking status and a charge level for the battery; an altitude element configured to receive inputs from a user to instruct the drone to increase or decrease altitude; a yaw element configured to receive inputs from a user to instruct the drone to roll left or roll right; a velocity element configured to receive inputs from a user to instruct the drone to increase or decrease speed; a deployment element configured to receive inputs from a user to instruct the drone to deploy from the docking station; and a return element configured to receive inputs from a user to instruct the drone to return to the docking station.
 13. The vehicle according to claim 1, wherein the interior includes a cargo bin having a movable lid to provide and prevent access to the cargo bin, the docking station being located in the cargo bin.
 14. The vehicle according to claim 1, wherein the vehicle body includes an opening that is configured to permit the drone to enter and exit the vehicle.
 15. The vehicle according to claim 14, wherein the opening is a sun roof or a moon roof.
 16. The vehicle according to claim 1, wherein the vehicle includes a removable roof, and when the roof is removed from the vehicle the drone is permitted to exit and enter the vehicle.
 17. A method for operating a drone using the vehicle of claim 1, comprising: receiving, by the vehicle ECU, a request to initiate the flight operations from either a graphic user interface (GUI) of the vehicle or a remote operating device; wherein, based on the request received by the vehicle ECU to initiate the flight operations, the ECU is configured to communicate with vehicle sensors to determine whether weather conditions are suitable to conduct the flight operations, and based on signals generated by the vehicle sensors that are indicative of the weather conditions, the vehicle ECU is configured to communicate instructions to the docking station controller either to permit the drone to conduct the flight operations or prevent the drone from conducting the flight operations.
 18. The method of claim 17, wherein after the vehicle ECU receives the request to initiate the flight operations, the method further comprises determining whether the vehicle is in motion and, based on a determination whether the vehicle is in motion, the vehicle ECU is configured to communicate instructions to the docking station controller either to permit the drone to conduct the flight operations or prevent the drone from conducting the flight operations.
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